How to read a paper (Part I)

Tao Du

The logic behind a scientific paper

- What is the research problem?
  - Is it an important problem, and is it of interest to the scientific community?
- What are the state of the arts (SOTAs)?
  - What are their limitations? Why is it necessary to develop a new method?
- What is the main method?
  - Describe the technical details behind it.
- What are the results?
  - Show evidence of the method outperforming SOTAs on certain metrics.

The logic behind a scientific paper

- What is the research problem? -> **Introduction**
  - Is it an important problem, and is it of interest to the scientific community?
- What are the state of the arts (SOTAs)? -> **Related Work**
  - What are their limitations? Why is it necessary to develop a new method?
- What is the main method? -> **Method**
  - Describe the technical details behind it.
- What are the results? -> **Experiments & Discussions**
  - Show evidence of the method outperforming SOTAs on certain metrics.

A recommended reading order

Abstract

-> **Intro** -> **Related Work**

-> **Experiments & Discussions**

-> Methods

-> Conclusions

How to read an abstract

Reading abstracts

Typical expectations for an abstract

- Describe the problem;
- Mention the weaknesses of related works;
- Explain the insights behind your approach;
- Report your experimental results.

3D Gaussian Splatting for Real-Time Radiance Field Rendering

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GEORGIOS KOPANAS*, Inria, Université Côte d'Azur, France
THOMAS LEIMKÜHLER, Max-Planck-Institut für Informatik, Germany
GEORGE DRETTAKIS, Inria, Université Côte d'Azur, France



Fig. 1. Our method achieves real-time rendering of radiance fields with quality that equals the previous method with the best quality [Barron et al. 2022], while only requiring optimization times competitive with the fastest previous methods [Fridovich-Keil and Yu et al. 2022; Müller et al. 2022]. Key to this performance is a novel 3D Gaussian scene representation coupled with a real-time differentiable renderer, which offers significant speedup to both scene optimization and novel view synthesis. Note that for comparable training times to InstantNGP [Müller et al. 2022], we achieve similar quality to theirs; while this is the maximum quality they reach, by training for 51min we achieve state-of-the-art quality, even slightly better than Mip-NeRF360 [Barron et al. 2022].

Radiance Field methods have recently revolutionized novel-view synthesis of scenes captured with multiple photos or videos. However, achieving high visual quality still requires neural networks that are costly to train and render, while recent faster methods inevitably trade off speed for quality. For unbounded and complete scenes (rather than isolated objects) and 1080p resolution rendering, no current method can achieve real-time display rates. We introduce three key elements that allow us to achieve state-of-the-art visual quality while maintaining competitive training times and importantly allow high-quality real-time (≥ 30 fps) novel-view synthesis at 1080p resolution. First, starting from sparse points produced during camera calibration, we represent the scene with 3D Gaussians that preserve desirable properties of continuous volumetric radiance fields for scene optimization while avoiding unnecessary computation in empty space; Second, we perform interleaved optimization/density control of the 3D Gaussians, notably optimizing anisotropic covariance to achieve an accurate representation of the scene; Third, we develop a fast visibility-aware rendering algorithm that supports anisotropic splatting and both accelerates training and allows real-time rendering. We demonstrate state-of-the-art visual quality and real-time rendering on several established datasets.

CCS Concepts: • **Computing methodologies** → **Rendering**; **Point-based models**; **Rasterization**; **Machine learning approaches**.

*Both authors contributed equally to the paper.

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Additional Key Words and Phrases: novel view synthesis, radiance fields, 3D gaussians, real-time rendering

ACM Reference Format:
Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2018. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Trans. Graph.* 0, 0, Article 0 (2018), 14 pages. https://doi.org/XXXXXXX.XXXXXXX

1 INTRODUCTION

Meshes and points are the most common 3D scene representations because they are explicit and are a good fit for fast GPU/CUDA-based rasterization. In contrast, recent Neural Radiance Field (NeRF) methods build on continuous scene representations, typically optimizing a Multi-Layer Perceptron (MLP) using volumetric ray-marching for novel-view synthesis of captured scenes. Similarly, the most efficient radiance field solutions to date build on continuous representations by interpolating values stored in, e.g., voxel [Fridovich-Keil and Yu et al. 2022] or hash [Müller et al. 2022] grids or points [Xu et al. 2022]. While the continuous nature of these methods helps optimization, the stochastic sampling required for rendering is costly and can result in noise. We introduce a new approach that combines the best of both worlds: our 3D Gaussian representation allows optimization with state-of-the-art (SOTA) visual quality and competitive training times, while our tile-based splatting solution ensures real-time rendering at SOTA quality for 1080p resolution on several previously published datasets [Barron et al. 2022; Hedman et al. 2018; Knapitsch et al. 2017] (see Fig. 1).

Our goal is to allow real-time rendering for scenes captured with multiple photos, and create the representations with optimization times as fast as the most efficient previous methods for typical real scenes. Recent methods achieve fast training [Fridovich-Keil

An illustrative example by Dr. Jingwei Huang

一种加辣椒的番茄炒蛋

摘要

本文提出了一种新型的加辣椒的番茄炒蛋烹饪法。其最重要的理念是要加辣椒。我们发现江苏的番茄炒蛋太甜，安徽的番茄炒蛋太咸，而本质上最好的解决方法就是加辣椒。从众多的辣椒中，我们从各种本性中进行分析，发现一个关键点是使用成熟度刚刚好的绿色的墨西哥辣椒。它由于个头小，不会喧宾夺主；由于绿色，增加了菜肴的颜色丰富度。最重要的是，这玩意真辣，谁吃谁知道。我们做了大量实验，发现老少咸宜，凡是吃了辣椒的都会快速吃完自己的米饭，可见这碗加辣椒的番茄炒蛋让人食欲大增。

Abstract explained

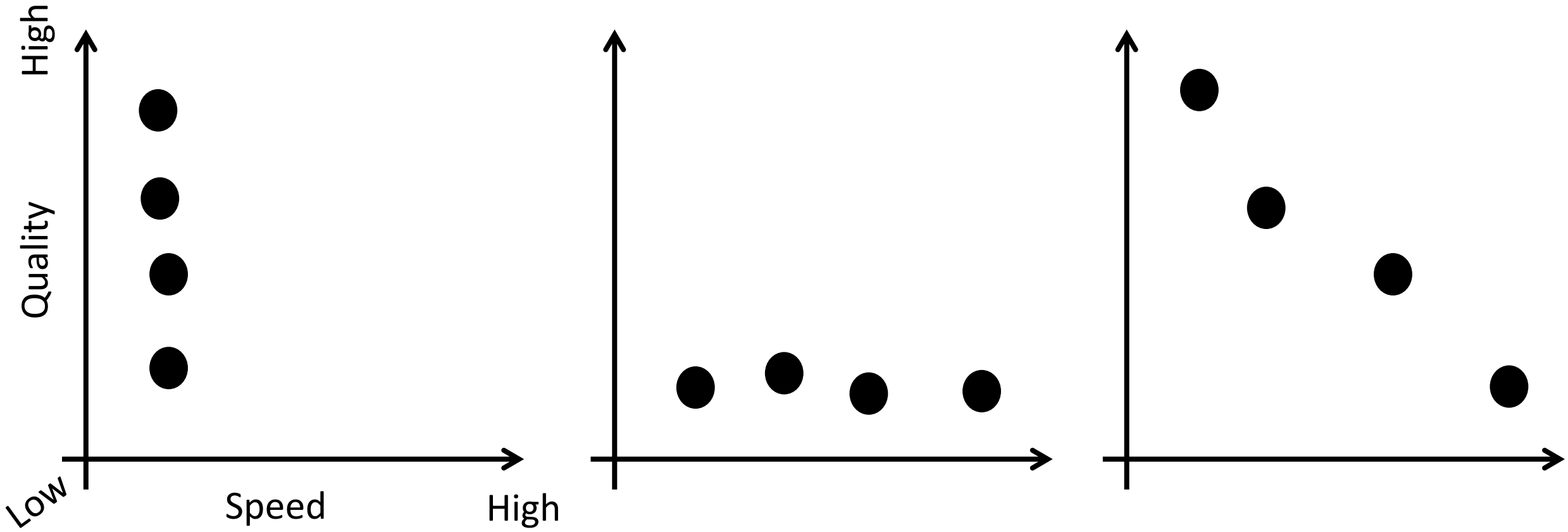
Problem	“番茄炒蛋烹饪法”
Related work	“江苏的番茄炒蛋太甜，安徽的番茄炒蛋太咸”
Method	“一个关键点是使用成熟度刚刚好的绿色的墨西哥辣椒”
Results	“我们做了大量实验，发现老少咸宜，凡是吃了辣椒的都会快速吃完自己的米饭”

Example: Gaussian Splatting (Para. one)

Radiance Field methods have recently revolutionized novel-view synthesis of scenes captured with multiple photos or videos. However, achieving high visual quality still requires neural networks that are costly to train and render, while recent faster methods inevitably trade off speed for quality. For unbounded and complete scenes (rather than isolated objects) and 1080p resolution rendering, no current method can achieve real-time display rates.

Quiz

Which figure best describes the existing methods?



Exercise: Gaussian Splatting (Para. one)

Radiance Field methods have recently revolutionized **novel-view synthesis** of scenes captured with multiple photos or videos. However, achieving **high visual quality** still requires neural networks that are **costly to train and render**, while recent **faster methods** inevitably **trade off speed for quality**. For unbounded and complete scenes (rather than isolated objects) and **1080p resolution rendering**, no current method can achieve **real-time display rates**.

Abstract explained

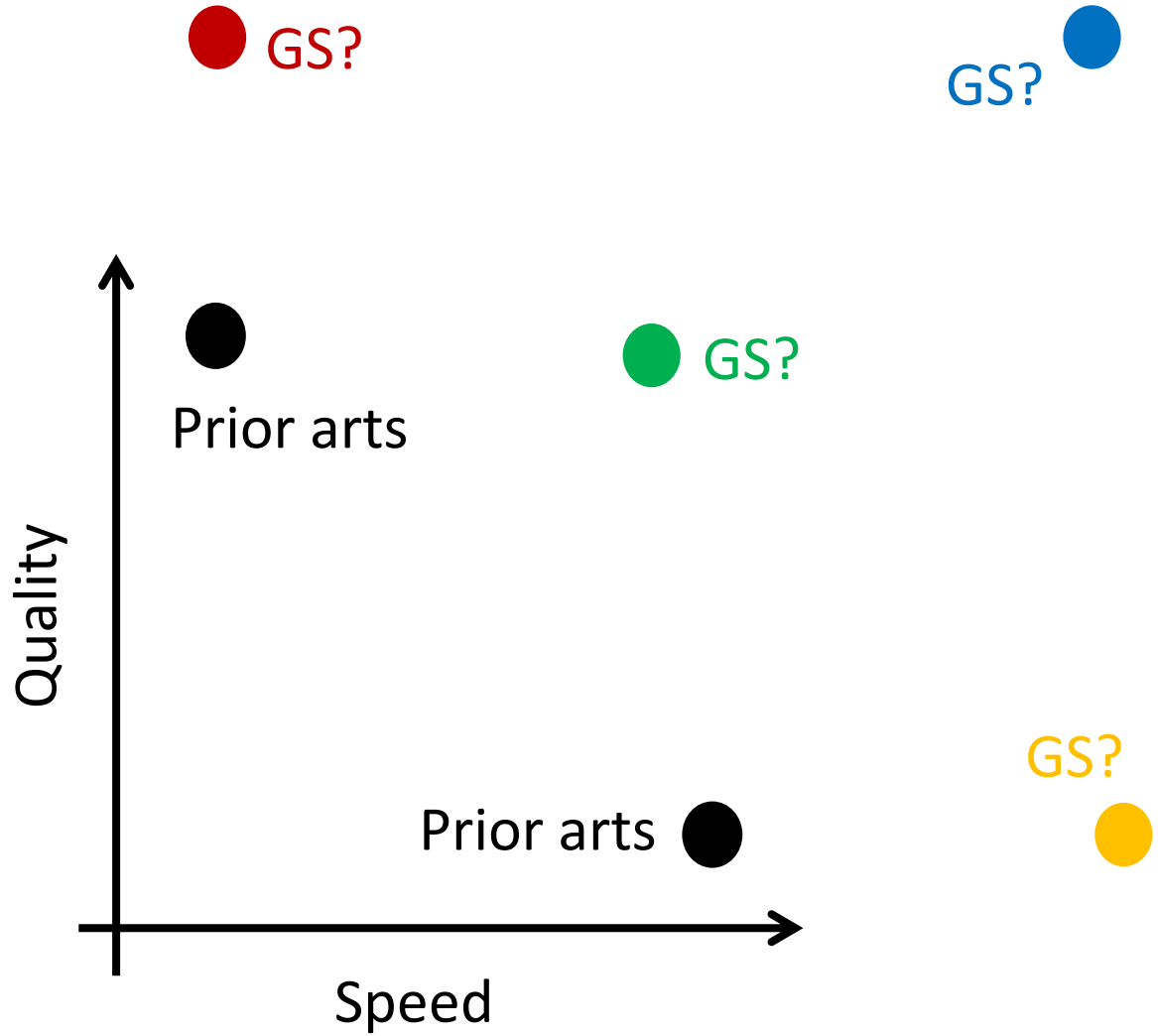
Problem	Novel-view synthesis
Related work	Cannot be both fast (real-time) and of high quality (1080p)
Method	
Results	

Example: Gaussian Splatting (Para. two)

We introduce three key elements that allow us to achieve state-of-the-art visual quality while maintaining competitive training times and importantly allow high-quality real-time (≥ 100 fps) novel-view synthesis at 1080p resolution.

Quiz

Which circle best describes the goal of Gaussian Splatting?



Exercise: Gaussian Splatting (Para. two)

We introduce three key elements that allow us to **achieve state-of-the-art visual quality** while **maintaining competitive training times** and importantly allow **high-quality real-time (≥ 100 fps)** novel-view synthesis at 1080p resolution.

Example: Gaussian Splatting (Para. three)

First, starting from sparse points produced during camera calibration, we represent the scene with 3D Gaussians that preserve desirable properties of continuous volumetric radiance fields for scene optimization while avoiding unnecessary computation in empty space; Second, we perform interleaved optimization/density control of the 3D Gaussians, notably optimizing anisotropic covariance to achieve an accurate representation of the scene; Third, we develop a fast visibility-aware rendering algorithm that supports anisotropic splatting and both accelerates training and allows realtime rendering. We demonstrate state-of-the-art visual quality and real-time rendering on several established datasets.

Example: Gaussian Splatting (Para. three)

First, starting from sparse points produced during camera calibration, we represent the scene with 3D Gaussians that preserve desirable properties of continuous volumetric radiance fields for scene optimization while avoiding unnecessary computation in empty space; **Second**, we perform interleaved optimization/density control of the 3D Gaussians, notably optimizing anisotropic covariance to achieve an accurate representation of the scene; **Third**, we develop a fast visibility-aware rendering algorithm that supports anisotropic splatting and both accelerates training and allows realtime rendering. We **demonstrate state-of-the-art visual quality** and **real-time rendering** on several established datasets.

Abstract explained

Problem	Novel-view synthesis
Related work	Cannot be both fast (real-time) and of high quality (1080p)
Method	Three key elements
Results	SOTA visual quality & real-time speed

How to read an introduction

Reading an introduction

Your goal: find information about

- Context of the research problem
- Challenges faced by existing methods
- Technical insights in the method
- Experimental results that support the claim

Example: Gaussian Splatting (Para. one)

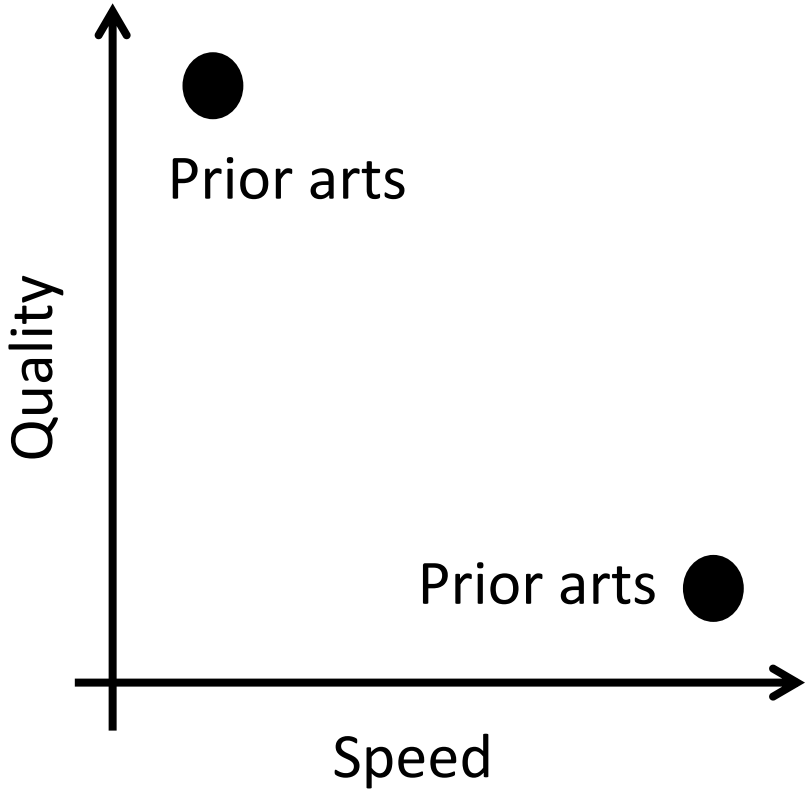
Meshes and points are the most common 3D scene representations because they are explicit and are a good fit for fast GPU/CUDA-based rasterization. In contrast, recent Neural Radiance Field (NeRF) methods build on continuous scene representations, typically optimizing a Multi-Layer Perceptron (MLP) using volumetric ray-marching for novel-view synthesis of captured scenes. Similarly, the most efficient radiance field solutions to date build on continuous representations by interpolating values stored in, e.g., voxel [Fridovich-Keil and Yu et al. 2022] or hash [Müller et al. 2022] grids or points [Xu et al. 2022]. While the continuous nature of these methods helps optimization, the stochastic sampling required for rendering is costly and can result in noise.

Example: Gaussian Splatting (Para. one)

Meshes and points are the most common 3D scene representations because they are **explicit and** are a **good** fit for **fast** GPU/CUDA-based rasterization. **In contrast,** recent Neural Radiance Field (**NeRF**) methods **build on continuous scene representations**, typically optimizing a Multi-Layer Perceptron (MLP) using volumetric ray-marching for novel-view synthesis of captured scenes. Similarly, the most efficient radiance field solutions to date build on continuous representations by interpolating values stored in, e.g., voxel [Fridovich-Keil and Yu et al. 2022] or hash [Müller et al. 2022] grids or points [Xu et al. 2022]. **While** the **continuous** nature of these **methods helps optimization**, the stochastic sampling required for rendering is **costly** and can **result in noise**.

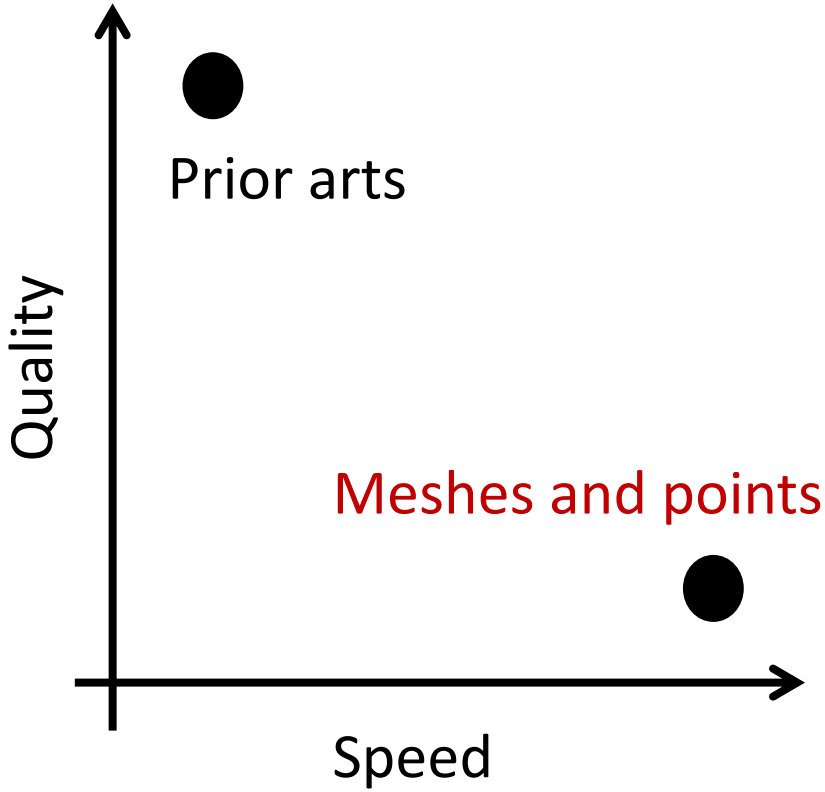
Recall what we learnt from its abstract

Fast methods exist;



Recall what we learnt from its abstract

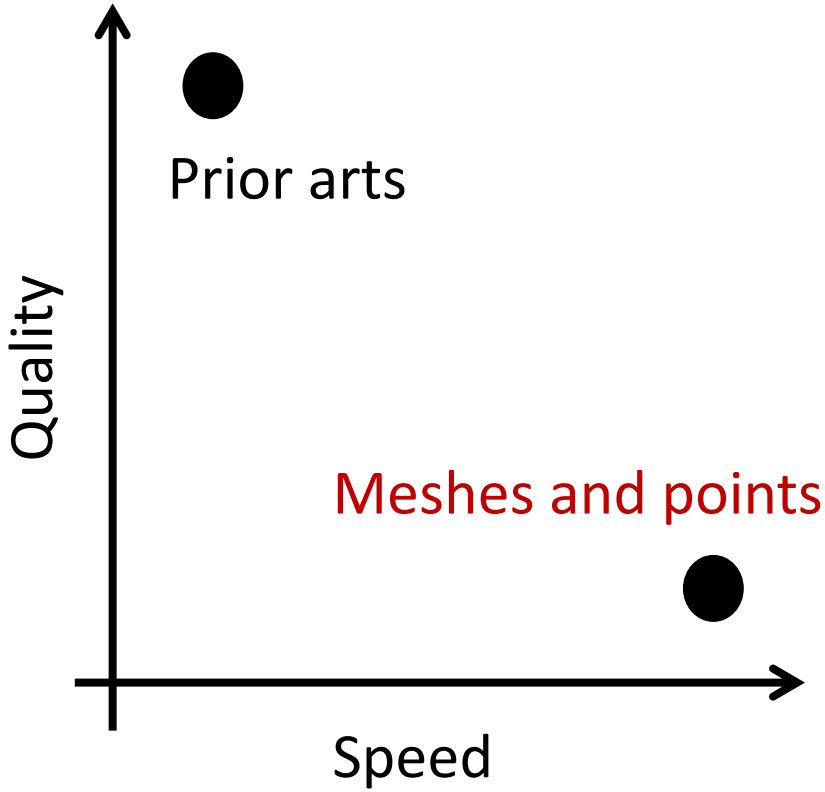
Fast methods exist -> Meshes and points



Recall what we learnt from its abstract

Fast methods exist -> Meshes and points

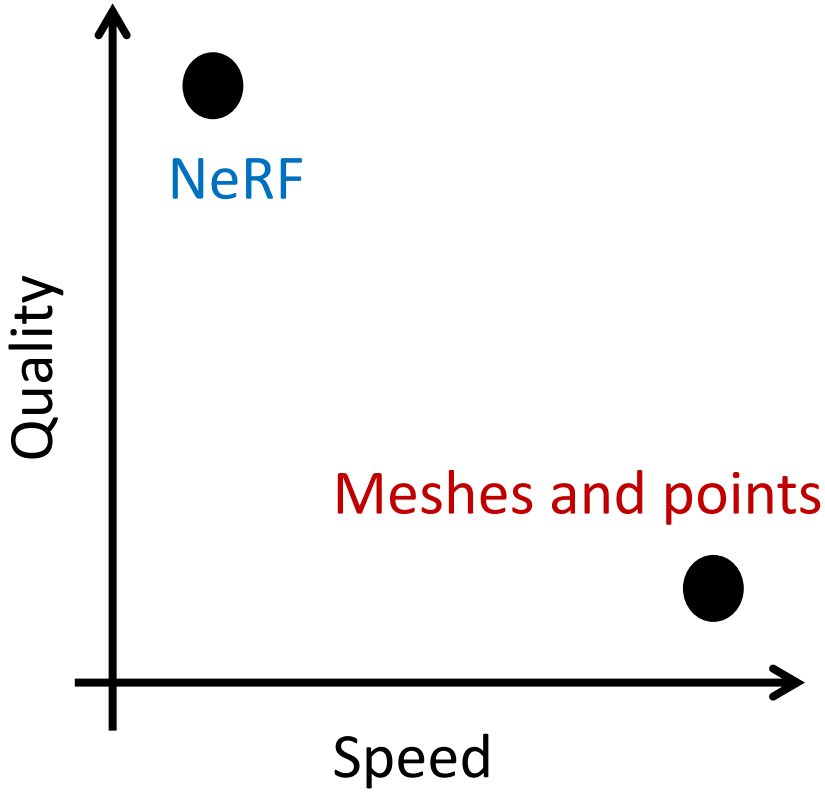
High-quality methods exist;



Recall what we learnt from its abstract

Fast methods exist -> Meshes and points

High-quality methods exist -> NeRF

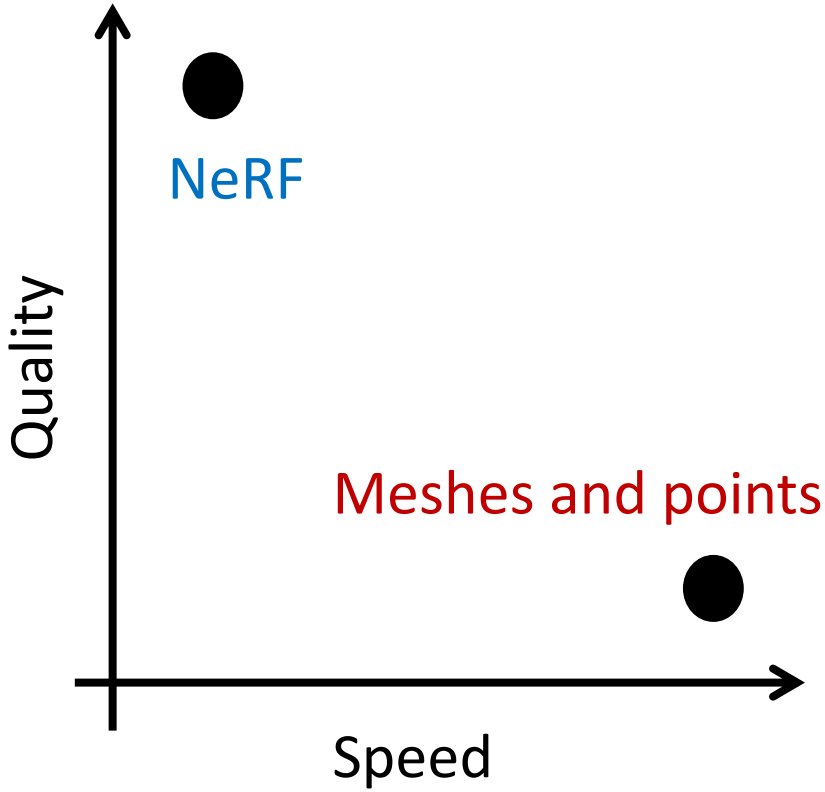


Recall what we learnt from its abstract

Fast methods exist -> Meshes and points

High-quality methods exist -> NeRF

No methods are both fast and of high-quality



The rest of the intro is left as an exercise

Read the full intro to find information about:

- Context of the research problem
- Challenges faced by existing methods
- Technical insights in the method
- Experimental results that support the claim

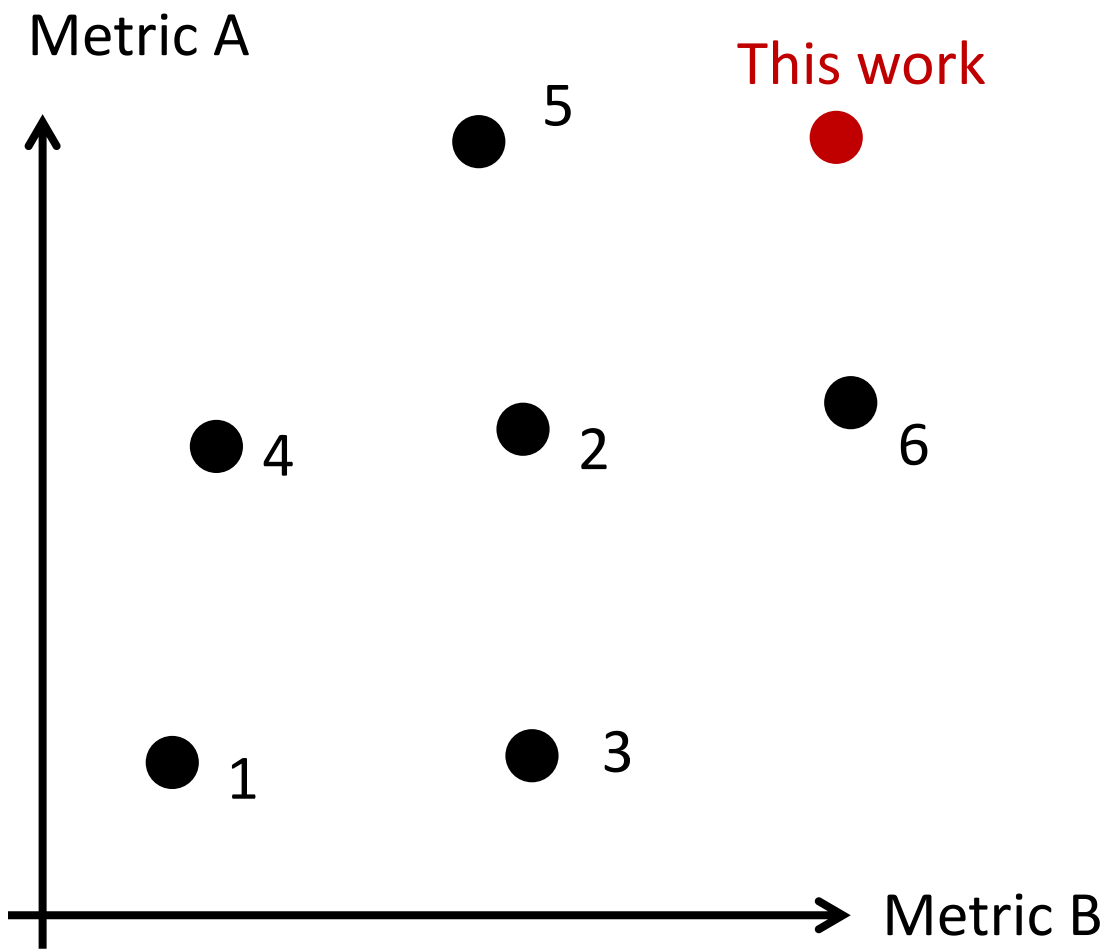
How to read related work

Reading related work

What makes a related work section easy to understand:

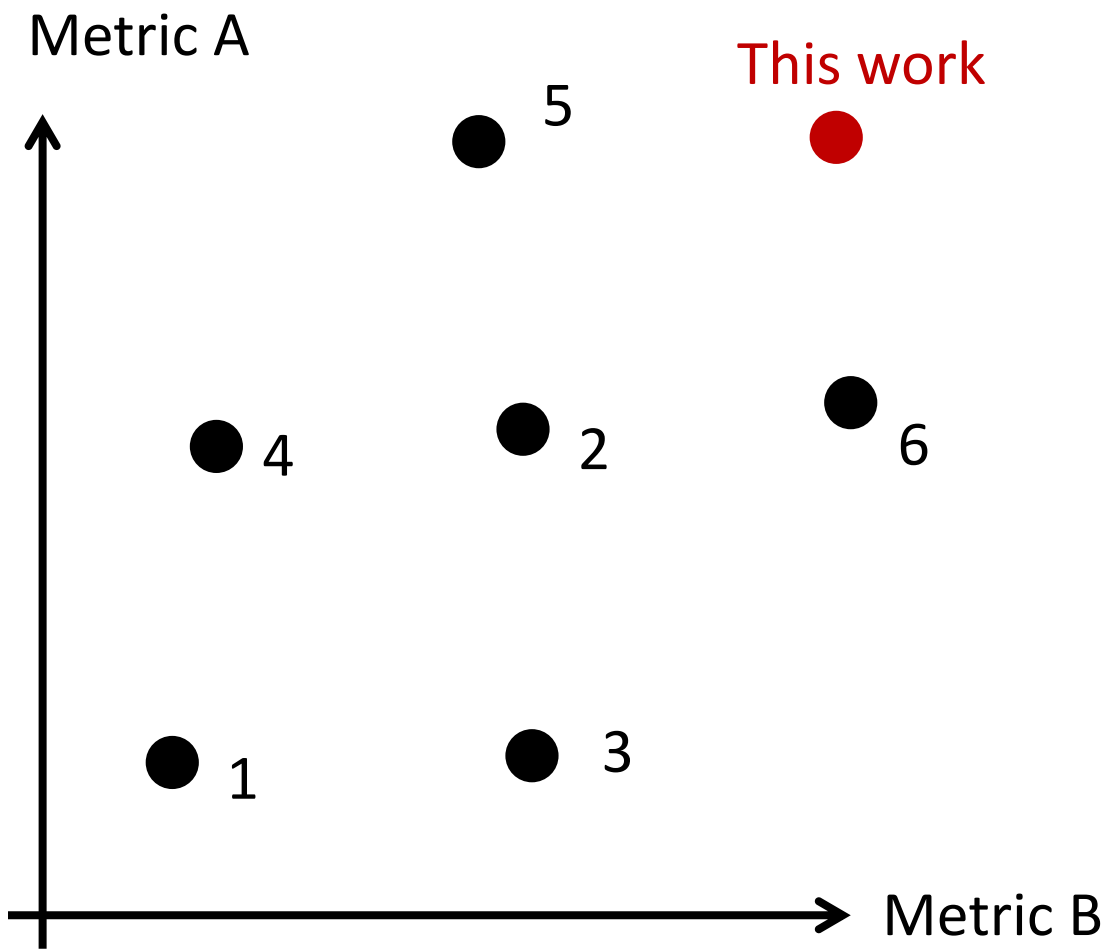
- A history of **milestone papers** and their **insights** into the problem;
- The **advantages** of this paper over them.

Reading related work



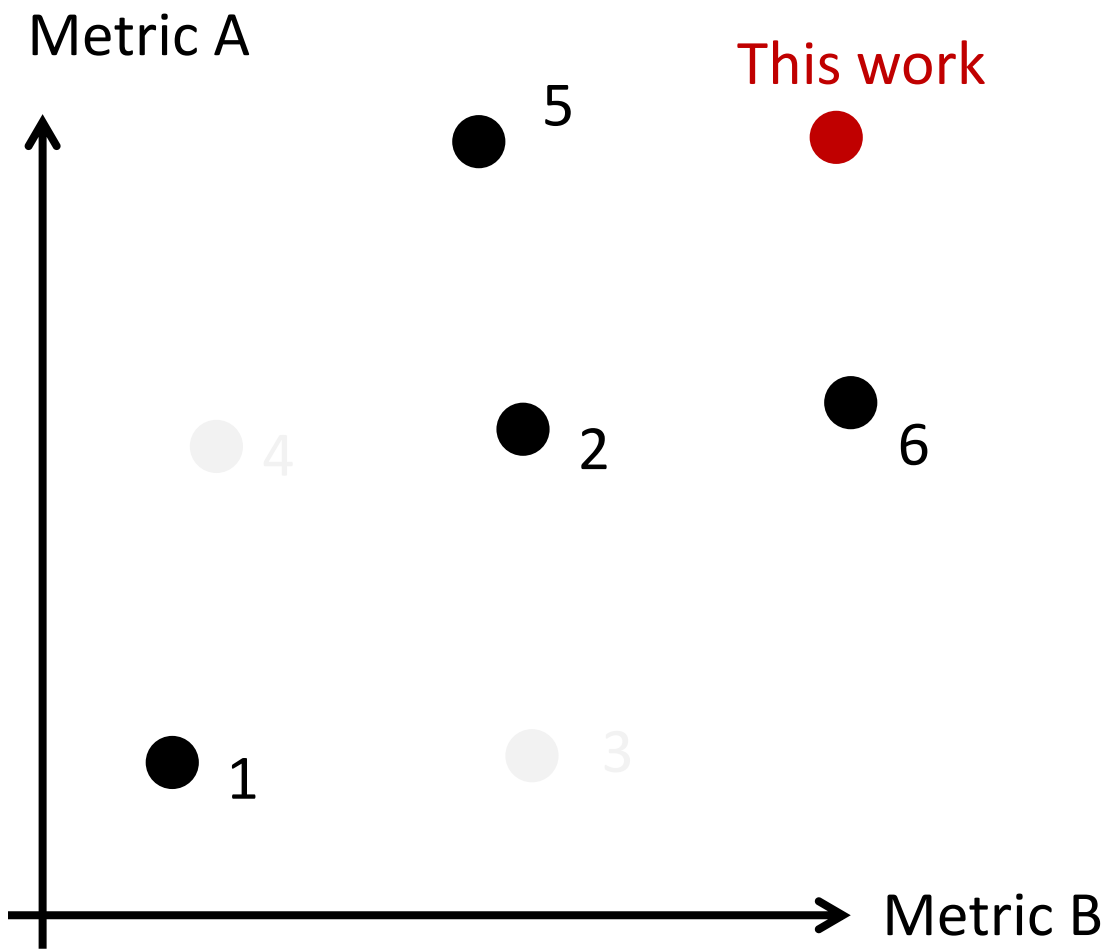
How to organize these papers in a related work section?

Reading related work



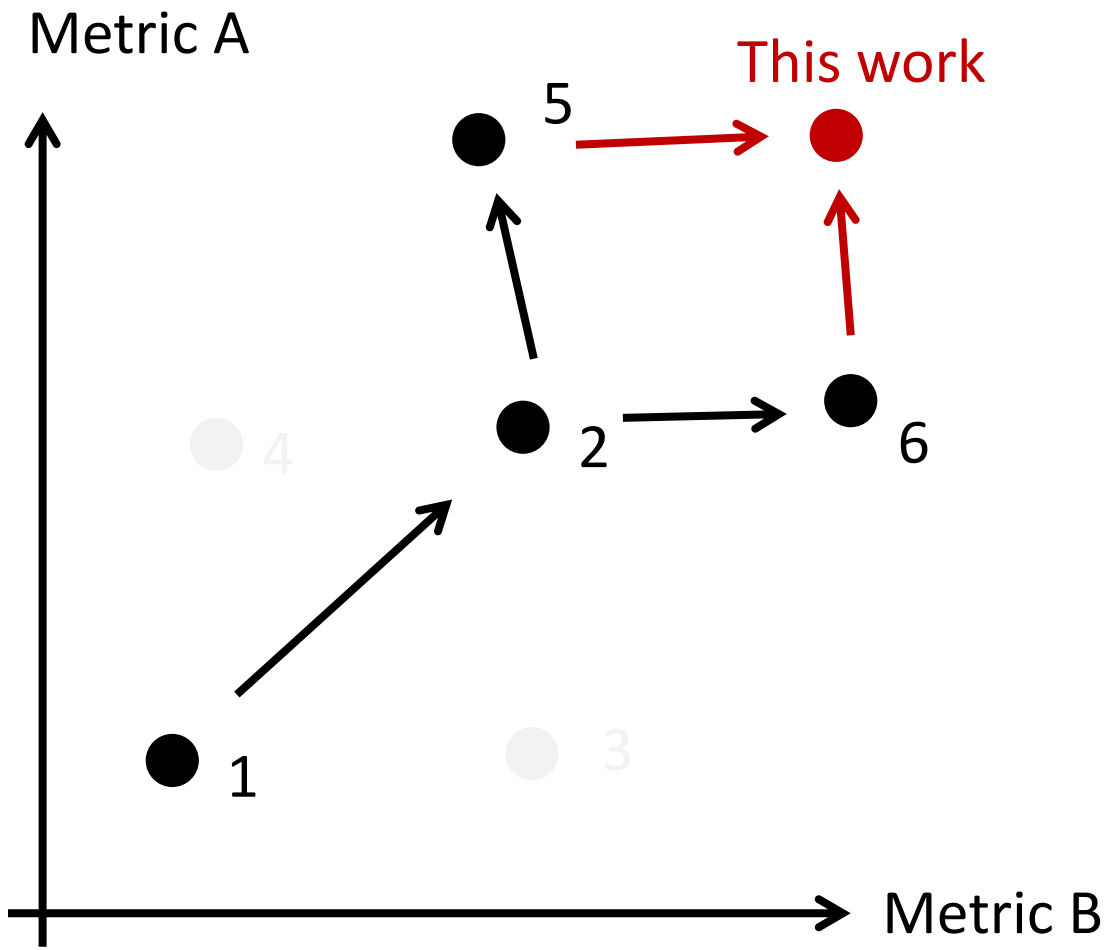
Proposal one (**chronological**):
1, 2, 3, 4, 5, 6, this work.

Reading related work



Proposal two (**chronological** + **Pareto front**):
1, 2, 5, 6, this work.

Reading related work



Proposal three (**chronological** + **Pareto front** + **relation**):

1 -> 2;

2 -> 5;

2 -> 6;

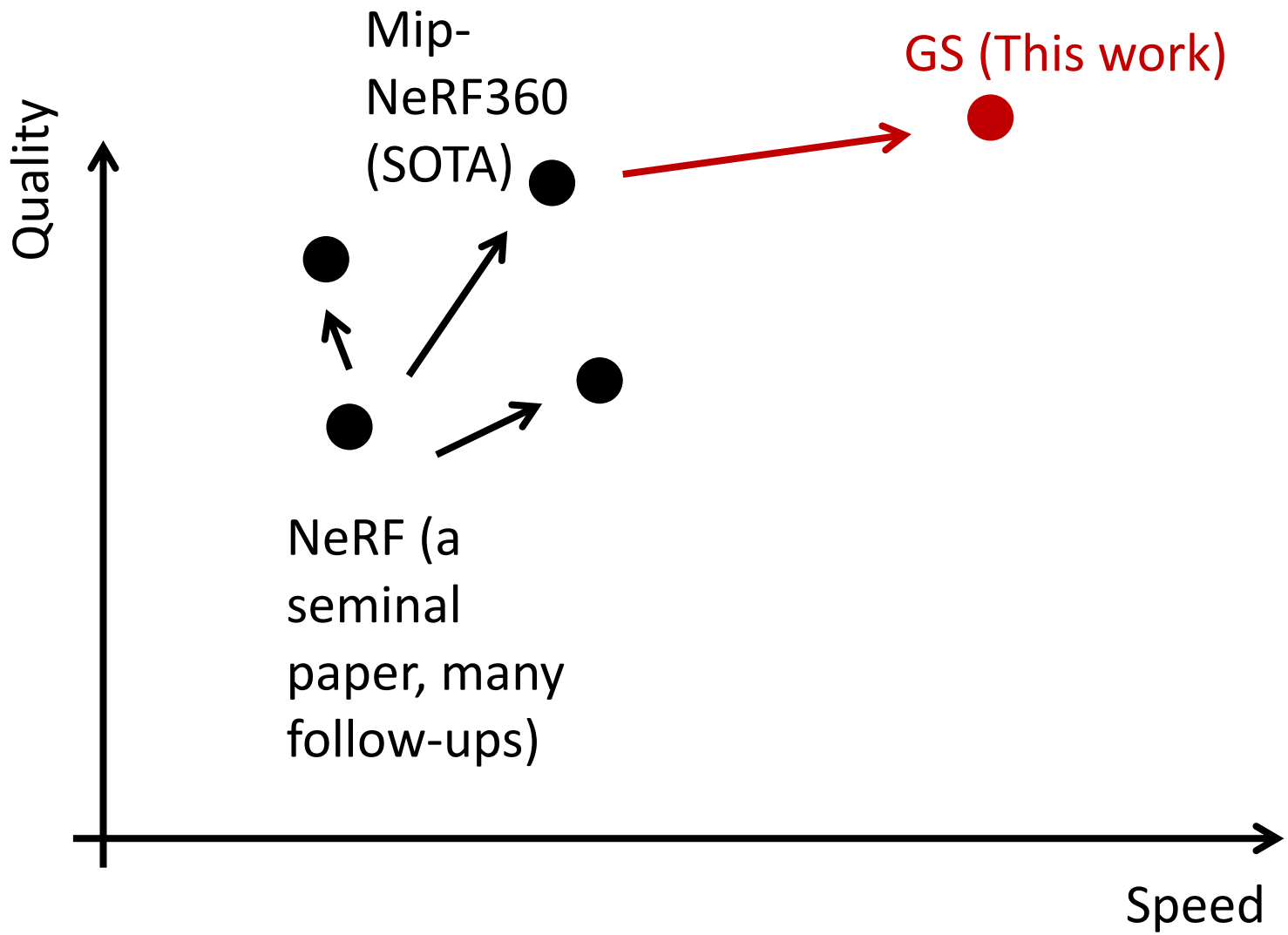
5 -> this work;

6 -> this work.

Example: Gaussian Splatting (Sec. 2.2)

…Neural Radiance Fields (NeRFs) [Mildenhall et al. 2020] introduced importance sampling and positional encoding to improve quality, but used a large Multi-Layer Perceptron negatively affecting speed. The success of NeRF has resulted in an explosion of follow-up methods that address quality and speed, often by introducing regularization strategies; the current state-of-the-art in image quality for novel-view synthesis is Mip-NeRF360 [Barron et al. 2022]. While the rendering quality is outstanding, training and rendering times remain extremely high; we are able to equal or in some cases surpass this quality while providing fast training and real-time rendering.

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Example: Gaussian Splatting (Sec. 2.2)

A history of **milestone papers** and their **insights** into the problem:

- NeRF
- Mip-NeRF360
- …

The **advantages** of this paper over them:

- Similar or better quality but faster speed